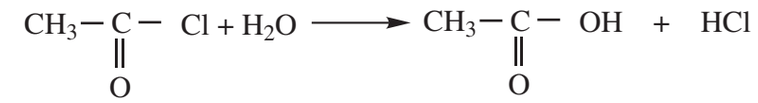
Formic acid cannot be prepared by Grignard reagent since the acid contains only one carbon atom

### 5. Hydrolysis of acylhalides and anhydrides

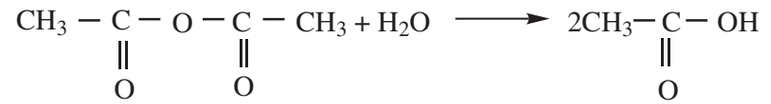
a) Acid chlorides when hydrolysed with water give Carboxylic acids.

**Example**

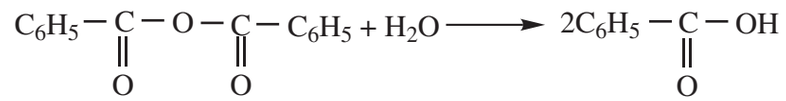


Acetyl chloride — Acetic acid

b) Acid anhydride when hydrolysed with water give corresponding carboxylic acids.



Acetic anhydride — Acetic Acid

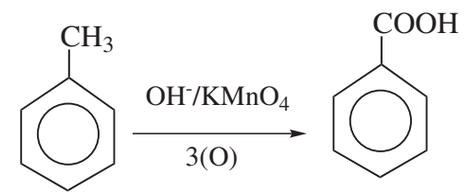


Benzoic anhydride — Benzoic acid

### 6. Oxidation of alkyl benzenes

Aromatic carboxylic acids can be prepared by vigorous oxidation of alkyl benzene with chromic acid or acidic or alkaline potassium permanganate. The entire side chain is oxidised to –COOH group irrespective of the length of the side chain.

**Example**



Toluene — Benzoic acid

**Evaluate yourself**

1) What happens when n-propyl benzene is oxidised using $H^+$ / $KMnO_4$?

2) How will you prepare benzoic acid using Grignard reagent.

## 12.11 Physical Properties of carboxylic acids.

i) Aliphatic carboxylic acid upto nine carbon atoms are colour less liquids with pungent odour. The higher members are odourless wax like solids.

ii) Carboxylic acids have higher boiling point than aldehydes, ketones and even alcohols of comparable molecular masses. This is due to more association of carboxylic acid molecules through intermolecular hydrogen bonding.



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